Theorem 4.1.15 Let V be a finite dimensional vector space. Let $A \subseteq \mathrm{End}(V)$ be a unital subalgebra. Set $B := \mathrm{Comm}(A)$. Then

(i) $\mathrm{Comm}(B) \supseteq A$

(ii) If V is a completely reducible A-module, then $\mathrm{Comm}(B) = A$.

Proof (i) is trivial. We prove (ii): need to show $\mathrm{Comm}(B) \subseteq A$.

Step 1 Fix a basis $\{v_1, \dots, v_n\}$ of V. It suffices to show: $\forall_{T \in \mathrm{Comm}(B)} \exists_{S \in A} \forall_{1 \le i \le n} Sv_i = Tv_i$.

Step 2 The module $V^{\oplus n}$ is completely reducible. Set $w_0 := v_1 \oplus \cdots \oplus v_n \in V^{\oplus n}$ and $W := A \cdot w_0 \subset V^{\oplus n}$.

Step 3 By Prop. 4.1.11, W has a complement in $V^{\oplus n}$. That is, there exists an A-submodule $W' \subseteq V^{\oplus n}$ such that $V^{\oplus n} = W \oplus W'$.

Step 4 Let $P : V^{\oplus n} \to W \subseteq V^{\oplus n}$ be the canonical projection. Recall the linear maps

$$P_{j,i} : V \xrightarrow[i\text{-th}]{} V^{\oplus n} \xrightarrow{P} V^{\oplus n} \xrightarrow[j\text{-th}]{} V .$$

Step 5 We have $T^{\oplus n} w_0 := (Tv_1, \dots, Tv_n) = T^{\oplus n} P w_0 = P T^{\oplus n} w_0$. Thus $T^{\oplus n} w_0 \in W$.

($P w_0 = w_0$ since; see proof of Jacobson density theorem)

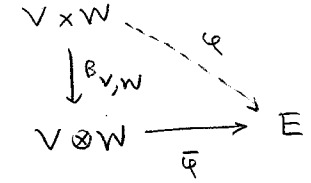
Step 6 By definition of W, Step 5 gives

$$(Tv_1, \dots, Tv_n) \in W \Rightarrow \exists_{S \in A} : S \cdot w_0 = (Tv_1, \dots, Tv_n).$$

Step 7 Thus $(Sv_1, \dots, Sv_n) = (Tv_1, \dots, Tv_n)$, that is, $Sv_i = Tv_i$ for all $1 \le i \le n$.

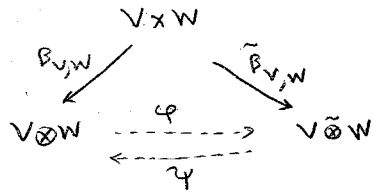
Tensor products of vector spaces

Def Let V, W be two vector spaces. The tensor product $V \otimes W$ is a vector space such that the exists a bilinear map $V \times W \xrightarrow{\beta_{V,W}} V \otimes W$ which satisfies the following universal property: Given any vector space E and a bilinear map $V \times W \xrightarrow{\varphi} E$, there exists a unique linear map $V \otimes W \xrightarrow{\bar{\varphi}} E$ such that $\bar{\varphi} \circ \beta_{V,W} = \varphi$



Proposition The tensor product of V and W, if exists, is unique.

proof Consider two candidates $V \otimes W$ and $V \tilde{\otimes} W$ for tensor product. The universal property implies existence of linear maps $V \otimes W \underset{\psi}{\overset{\varphi}{\rightleftarrows}} V \tilde{\otimes} W$ such that



$$\varphi \circ \beta_{V,W} = \tilde{\beta}_{V,W} \quad \text{and} \quad \psi \circ \tilde{\beta}_{V,W} = \beta_{V,W}$$

In particular, $\varphi \circ \psi \circ \tilde{\beta}_{V,W} = \varphi \circ \psi$ & $\psi \circ \varphi \circ \beta_{V,W} = \psi \circ \varphi$.

Uniqueness now implies $\varphi \circ \psi = 1_{V \tilde{\otimes} W}$ & $\psi \circ \varphi = 1_{V \otimes W}$.

Example $V \simeq \mathbb{C}^m$, $W = \mathbb{C}^n$. We choose bases $v_1, \dots, v_m$ of V and $w_1, \dots, w_n$ of W.

Consider the vector space $U \simeq \mathbb{C}^{mn}$ with a basis indexed as $u_{ij} := v_i \otimes w_j$.

$$V \times W \longrightarrow U$$
$$\sum a_i v_i \times \sum b_j w_j \longmapsto \sum a_i b_j u_{ij}$$

Given any bilinear map $V \times W \xrightarrow{\varphi} E$, the linear map $\bar{\varphi} : U \longrightarrow E$ is unique because it should satisfy $\bar{\varphi}(u_{ij}) = \bar{\varphi}(v_i \otimes w_j) = \varphi(v_i \times w_j)$.

Moreover, the latter relation indeed defines $\bar{\varphi}$ uniquely (here $\bar{\varphi}$ exists!).

Remark Let $\mathcal{A}$ and $\mathcal{B}$ be two associative algebras. Then $\mathcal{A} \otimes \mathcal{B}$ is an associative algebra with product $(a \otimes b)(a' \otimes b') := aa' \otimes bb'$.

Let (ρ, V) be an $\mathcal{A}$-module and let (σ, W) be a $\mathcal{B}$-module. Then $(\rho \otimes \sigma, V \otimes W)$ is an $\mathcal{A} \otimes \mathcal{B}$-module defined by $\rho \otimes \sigma (a \otimes b)(v \otimes w) = \rho(a) v \otimes \sigma(b) w$.

Lemma Assume that (ρ, V) is an irreducible $\mathcal{A}$-module and (σ, W) is an irreducible $\mathcal{B}$-module. Further assume that $\dim V < \infty$ & $\dim W < \infty$. Then $(\rho \otimes \sigma, V \otimes W)$ is an irreducible $\mathcal{A} \otimes \mathcal{B}$-module. (we assume that $\mathcal{A}$ & $\mathcal{B}$ are unital).

Proof Clearly $\mathrm{Im}(\rho \otimes \sigma) = \mathrm{Im}(\rho) \otimes \mathrm{Im}(\sigma) \subseteq \mathrm{End}(V \otimes W)$. But $\mathrm{Im}(\rho) = \mathrm{End}(V)$ & $\mathrm{Im}(\sigma) = \mathrm{End}(W)$ (follow from Corollary 4.1.7). Thus $\mathrm{Im}(\rho \otimes \sigma) = \mathrm{End}(V) \otimes \mathrm{End}(W) = \mathrm{End}(V \otimes W)$.

Isotypic decomposition and multiplicities

Let A be a unital associative algebra.

Def Let U be a finite dimensional irreducible A-module. We denote the class of A-modules equivalent to U by $[U]$.

We use $\hat{A}$ to denote the set of equivalence classes $[U]$ (U: finite dim'l irreducible A-module)

Given any A-module V, for every $\lambda \in \hat{A}$ we define the **λ-isotypic** subspace $V_{(\lambda)}$ by

$$V_{(\lambda)} := \sum_{U \subseteq V,\ [U]=\lambda} U .$$

Fix F_λ in the class λ for every $\lambda \in \hat{A}$. We define a map

$$S_\lambda : \mathrm{Hom}_A(F_\lambda, V) \otimes F_\lambda \longrightarrow V \qquad T \otimes w_\lambda \mapsto T w_\lambda$$

Def We define an A-module structure on $\mathrm{Hom}_A(F_\lambda, V) \otimes F_\lambda$: $a \cdot (T \otimes w) := T \otimes aw$. $a \in A$, $w \in F_\lambda$, $T \in \mathrm{Hom}(F_\lambda, V)$.

Def An A-module V is called **locally completely reducible** if for every $v \in V$, the submodule Av is completely reducible & finite dimensional.

Example $V = \mathbb{C}^2 \oplus \mathbb{C}^2 \oplus \cdots$ as a module for $\mathrm{Mat}_{2\times 2}(\mathbb{C})$. Note that the vectors $v \in V$ are of the form $(v_1, v_2, \ldots, v_N, 0, 0, \ldots)$ for $v_1, \ldots, v_N \in \mathbb{C}^2$.

Proposition 4.1.15 (i) The map S_λ is a homomorphism of A-modules and $\mathrm{im}(S_\lambda) = V_{(\lambda)}$.

(ii) If V is locally completely reducible then S_λ is an isomorphism onto $V_{(\lambda)}$.

Furthermore, $V \simeq \bigoplus_{\lambda \in \hat{A}} V_{(\lambda)}$.